

VERSION HISTORY

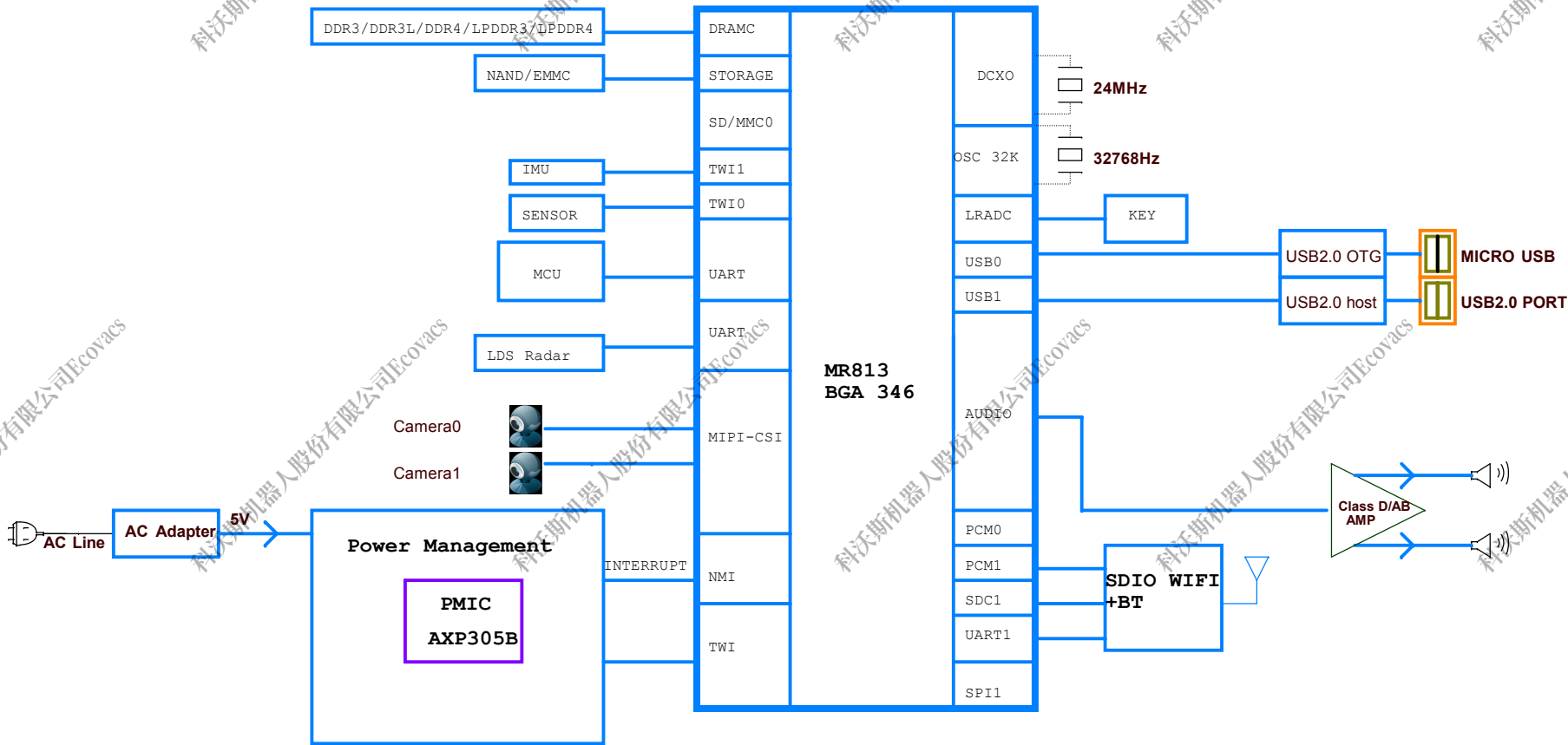
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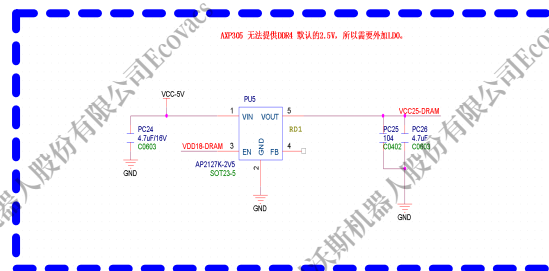
- P01 VERSION HISTORY
- P02 BLOCK DIAGRAM
- P03 POWER TREE
- P04 GPIO ASSIGNMENT
- P05 POWER
- P06 SOC-SYS
- P07 SOC-GPIO
- P08 DDR3 16X2
- P09 FLASH
- P10 AUDIO
- P11 USB/DEBUG
- P12 CAMREA
- P13 SENSORS/MCU/KEY
- P14 WIFI+BT

- P15 DDR4 16X2 4L V14

Revision	Description	Date	Drawn	Checked	Approved
Ver 0.1	Releas version	20200202			
Ver 0.2	P08 change to DDR3 16X2 only P05 Cancel DCDC6 , modify SWOUT to VCC-WIFI.	20200210			
Ver 0.3	1、update REF NAME and PAGEOFF number. 2、cancel 2 mipi connectors.	20200212			
Ver 0.9	1、HPOUTFB Connect GND 2、WREQIN CONNECT GND 3、WIFI CRYSTAL modify 40mhz	20200217			
Ver 1.0	1、add GPADC1 for DDR VERSION (reserved)	20200317			
Ver 1.1	1、CHANGE PMIC TO AXP305 2、ADD OVP CYCLE	20200327			
Ver 1.2	1、CHANGE DDR3 16X2 demo,cut some Caps. 2、Change VCC-RTC to VINT.	20200520			
Ver 1.3	1、CHANGE PMU(name) to AXP305B 2、Change VCC-RTC to BLD01. 3、ADD DDR4 16X2 cycle	20200706			
Ver 1.4	1 Add Vint Pull down resistor RP49.	20210311			
Ver 1.5	1 Update ddr4 REF	20220119			
Ver 1.6	Change OVP protect voltage from 6.5V to 7.2V	20230404			

BLOCK



☐ DEFAULT POWER OFF

GPIO ASSIGNMENT

PIN	Define	CFG	Function
PB0	CPUX_TMS	4	DEBUG
PB1	CPUX_TCK	4	
PB2	CPUX_TDO	4	
PB3	CPUX_TDI	4	
PB4		3	I2S
PB5		3	
PB6		3	
PB7		3	
PB8		3	DEBUG
PB9	CPUX_TX	2	
PB10	CPUX_RX	2	

PIN	Define	CFG	Function
PE0	MCSI_MCLK0	2	CSI
PE1	MCSI_SCK0	2	
PE2	MCSI_SDA0	2	
PE3	MCSI_SCK1	2	
PE4	MCSI_SDA1	2	
PE5	MCSI_MCLK1	2	
PE6	MCSIB_STBY_F	1	
PE7	MCSIB_RST_F	1	
PE8	MCSIA_STBY_R	1	
PE9	MCSIA_RST_R	1	

PIN	Define	CFG	Function
PH0	TWI0_SCK	2	TWI
PH1	TWI0_SDA	2	
PH2	TWI1_SCK	2	
PH3	TWI1_SDA	2	
PH4	SPI1-CS		GPIO
PH5	SPI1-CLK		
PH6	SPI1-MOSI		
PH7	SPI1-MISO		
PH8	USB0_ID_SOC	0	
PH9	USB0_DRVVBUS	1	
PH10	USB1_DRVVBUS	1	
PH11	IMU_INT	0	
PH12			
PH13			
PH14			
PH15			
PH16			
PH17			
PH18	PA-SHDN	1	
PH19	RADAR-PWREN	1	

PIN	Define	CFG	Function
PC0	NAND_WE/SDC2_DS	2/3	NAND/eMMC
PC1	NAND_ALE/SDC2_RST	2/3	
PC2	NAND_CLE	2	
PC3	NAND_CE1	2	
PC4	NAND_CE0	2	
PC5	NAND_RE/SDC2_CLK	2/3	
PC6	NAND_RB0/SDC2_CMD	2/3	
PC7	NAND_RB1	2	
PC8	NAND_DQ7/SDC2_D3	2/3	
PC9	NAND_DQ6/SDC2_D4	2/3	
PC10	NAND_DQ5/SDC2_D0	2/3	
PC11	NAND_DQ4/SDC2_D5	2/3	
PC12	NAND_DQS	2	
PC13	NAND_DQ3/SDC2_D1	2/3	
PC14	NAND_DQ2/SDC2_D6	2/3	
PC15	NAND_DQ1/SDC2_D2	2/3	
PC16	NAND_DQ0/SDC2_D7	2/3	

PIN	Define	CFG	Function
PF0	SDC0_D1	2	CARD
PF1	SDC0_D0	2	
PF2	SDC0_CLK	2	
PF3	SDC0_CMD	2	
PF4	SDC0_D3	2	
PF5	SDC0_D2	2	
PF6	SDC0_DET	0	

PIN	Define	CFG	Function
PL0	PMU_SCK	2	CPUS
PL1	PMU_SDA	2	
PL2	BT_RST_N	1	
PL3	BT_WAKE_AP	0	
PL4	AP_WAKE_BT	1	
PL5	WL_PMU_EN	1	
PL6	WL_WAKE_AP	0	
PL7	WL_PWR_ON	1	
PL8	S_TWI_SCK	2	
PL9	S_TWI_SDA	2	
PL10			
PL11	IR_IN	3	

PIN	Define	CFG	Function
PD0			LCD
PD1			
PD2			
PD3			
PD4			
PD5			
PD6			
PD7			
PD8			
PD9			
PD10			
PD11			
PD12			
PD13			UART3-TX
PD14	UART3-TX		
PD15	UART3-RX		
PD16	UART3-RTS		
PD17	UART3-CTS		
PD18	UART4-TX		
PD19	UART4-RX		
PD20	UART4-RTS		
PD21	UART4-CTS		
PD22	PWM1		PWM0
PD23	PWM0		

PIN	Define	CFG	Function
PG0	WL_SDIO_CLK	2	WIFI/BT
PG1	WL_SDIO_CMD	2	
PG2	WL_SDIO_D0	2	
PG3	WL_SDIO_D1	2	
PG4	WL_SDIO_D2	2	
PG5	WL_SDIO_D3	2	
PG6	BT_UART_RX	2	
PG7	BT_UART_TX	2	
PG8	BT_UART_CTS	2	
PG9	BT_UART_RTS	2	
PG10	BT_PCM_CLK	3	
PG11	BT_PCM_SYNC	3	
PG12	BT_PCM_DIN	3	
PG13	BT_PCM_DOUT	3	



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Design Name **MR813_DDR3_16X2**

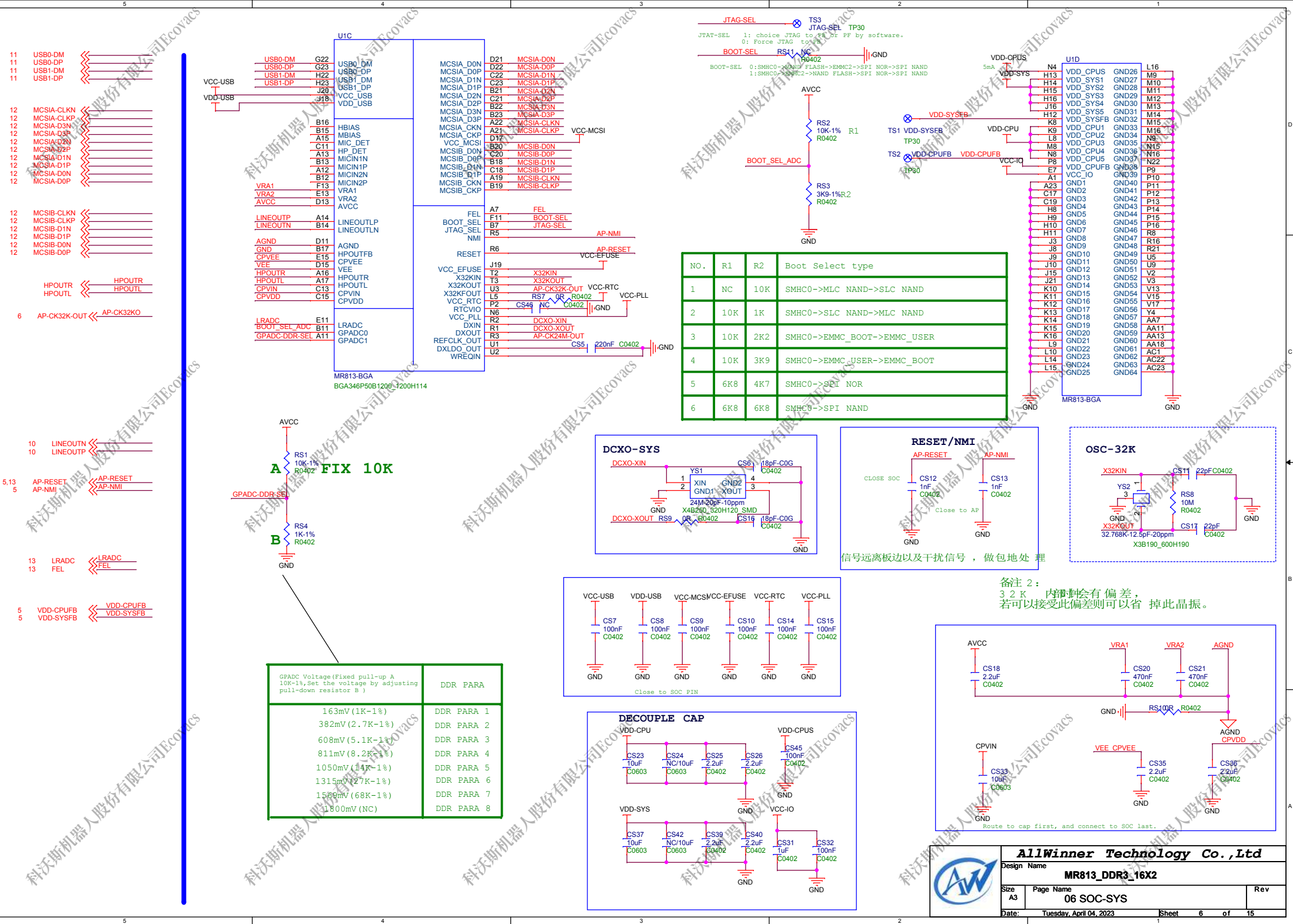
Size A3 Page Name **04 GPIO ASSIGNMENT** Rev

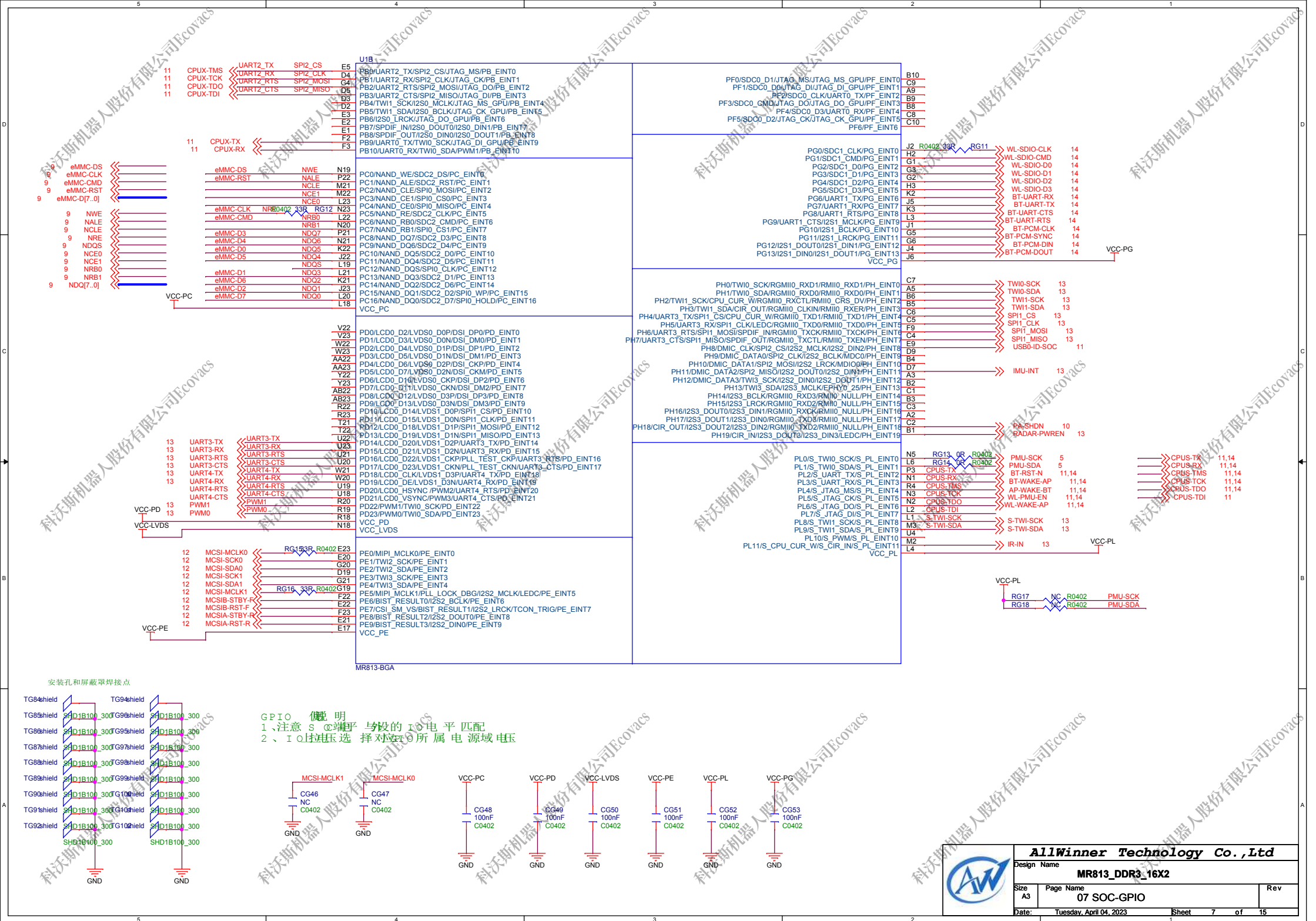
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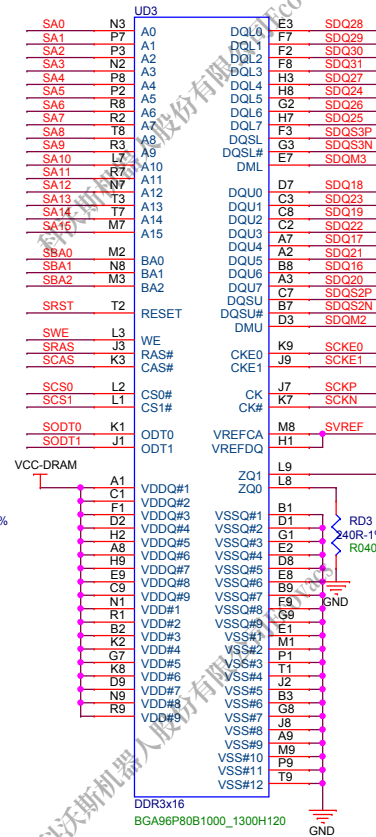
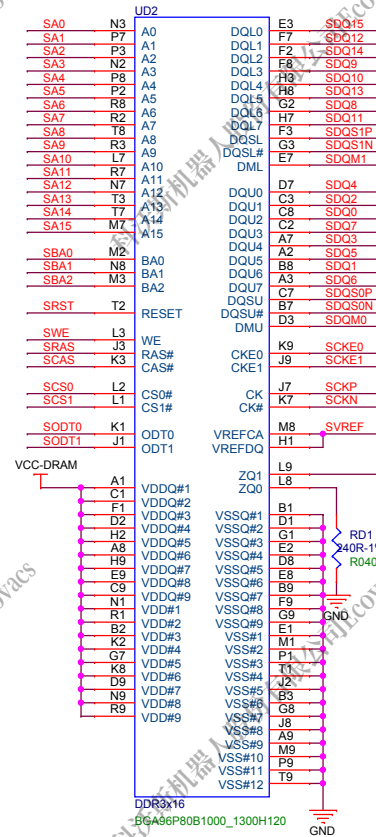
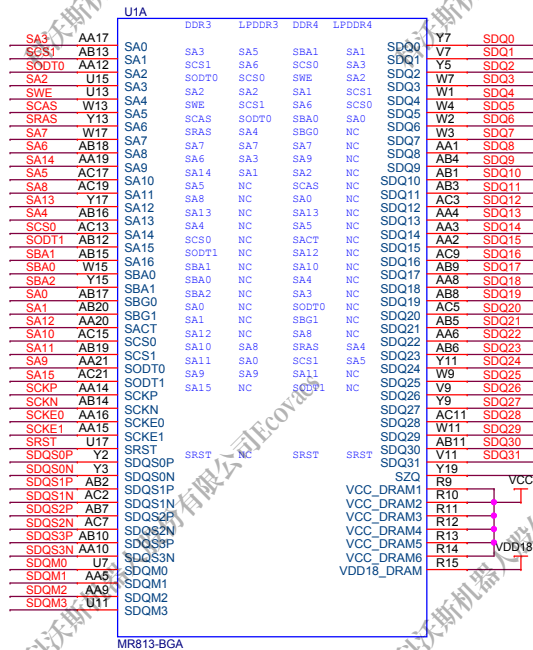
Design Name	MR813 DDR3 16X2
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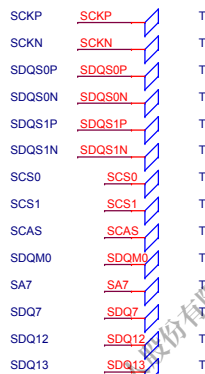




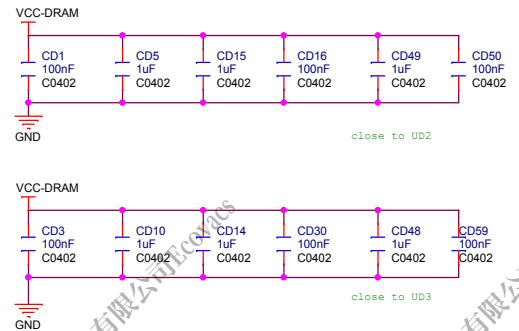
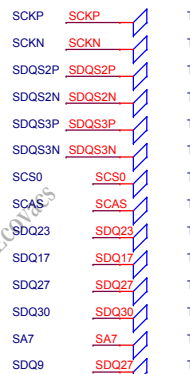
DDR3 16x2



close to UD2



close to UD3

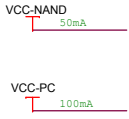
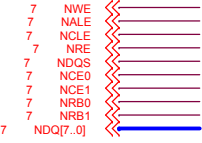
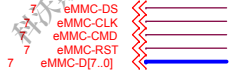


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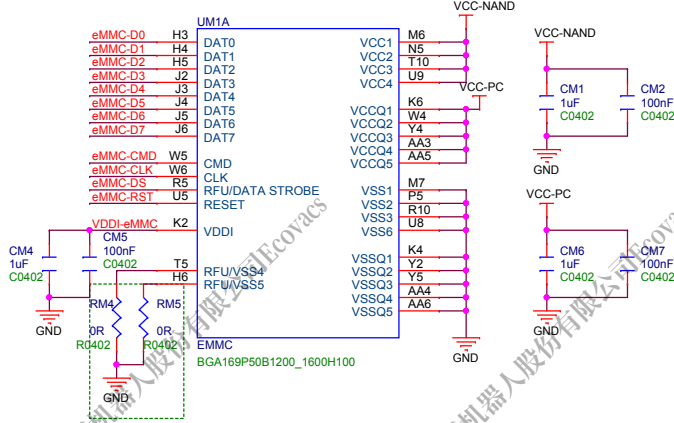
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Size: A3 Page Name: 08 DDR3 16X2 Rev: 1

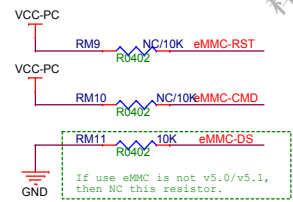
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eMMC

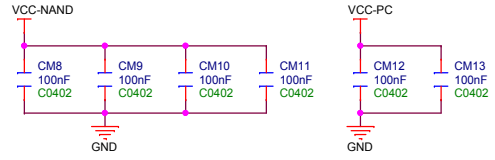
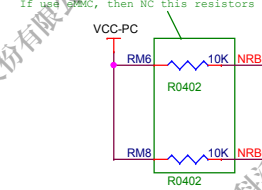
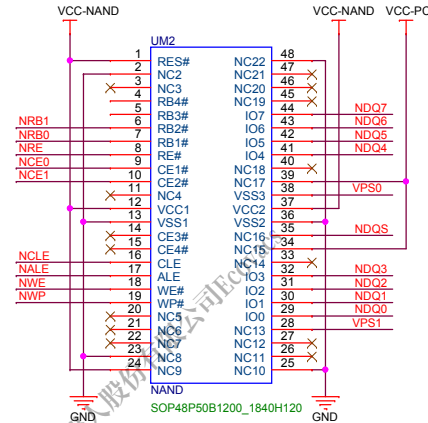


If eMMC is not v5.0/v5.1, then NC this two resistors.



NOTE:
THE VCC-PC VOLTAGE MATCH THE FLASH OR EMMC

NAND



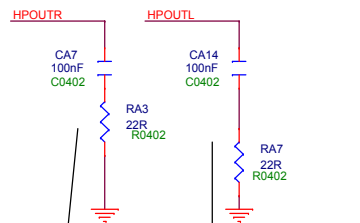
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	09 NAND/EMMC	
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HPOUTR
HPOUTL

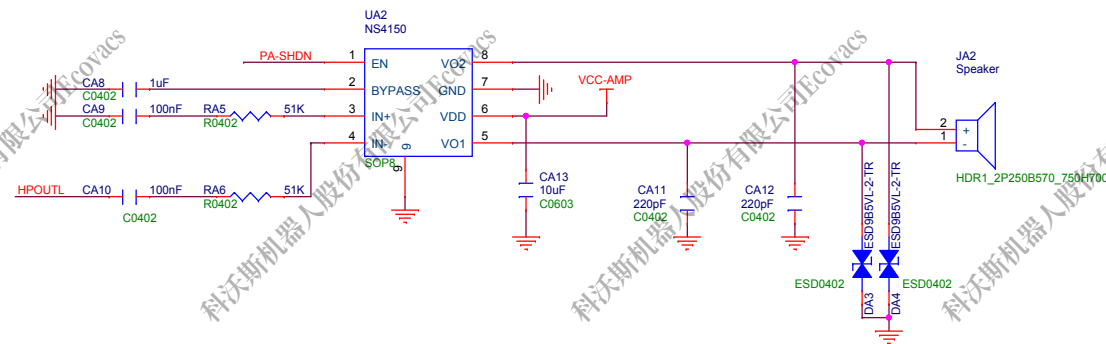
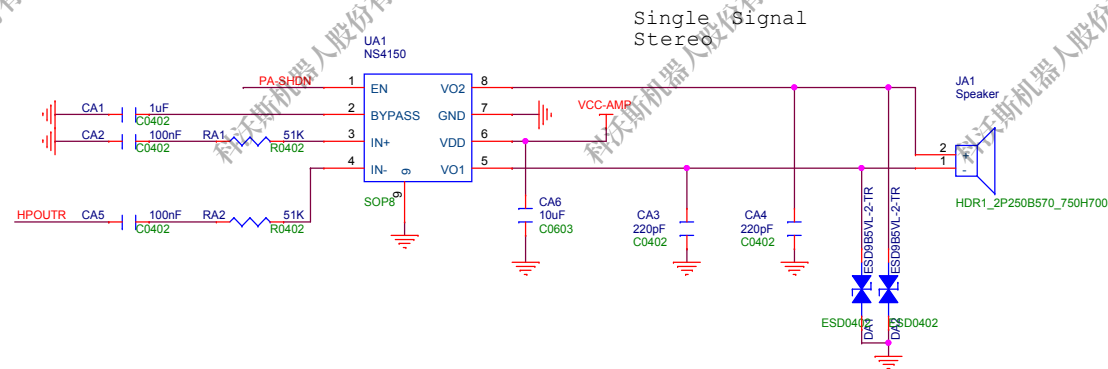
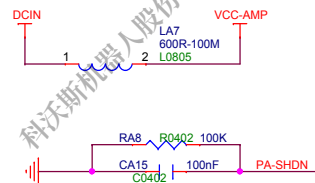
6 LINEOUTN
6 LINEOUTP

7 PA-SHDN

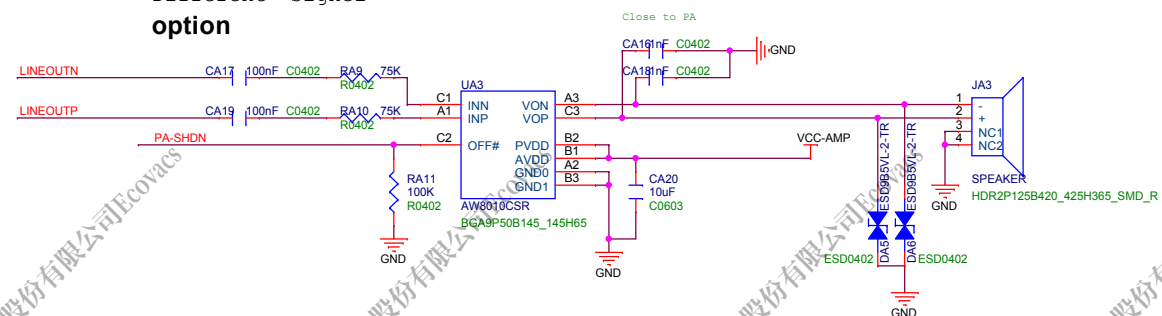
PS
1A



If use HPOUT, the RC must be reserved

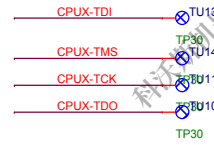
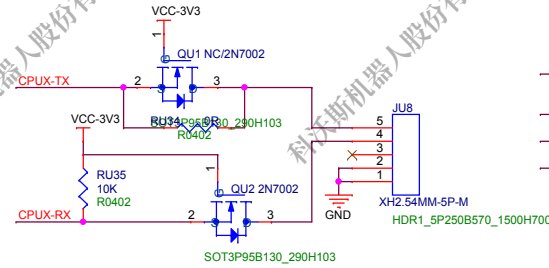


Different signal option



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	10 AUDIO	
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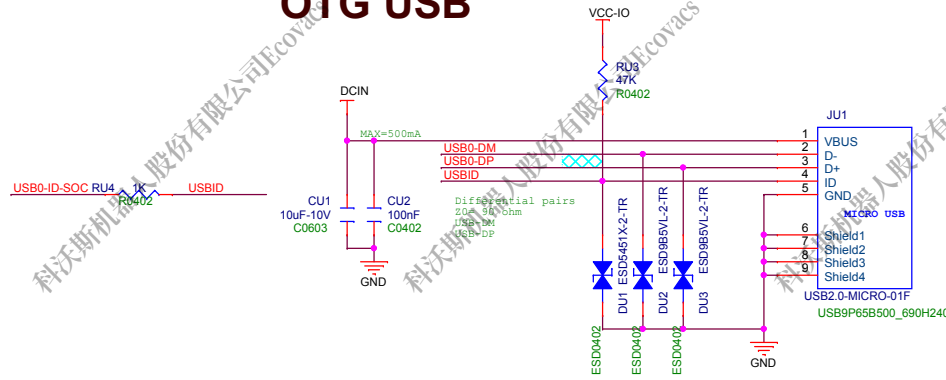
CPUX DEBUG



CPUS DEBUG

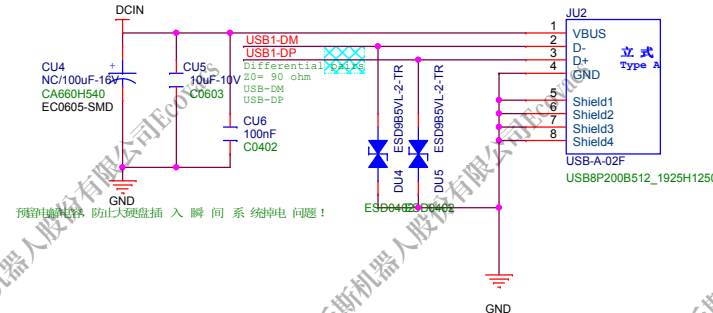


OTG USB



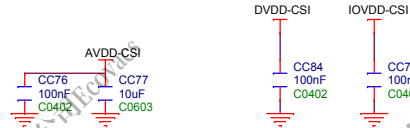
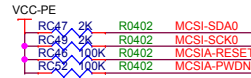
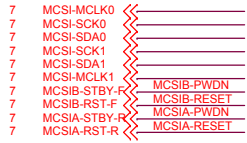
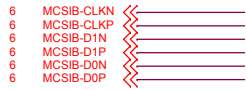
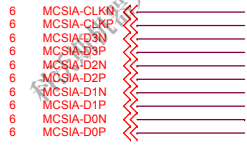
note: Make sure the routing between the ESD and the USB connectors should be on the same PCB side

HOST A USB

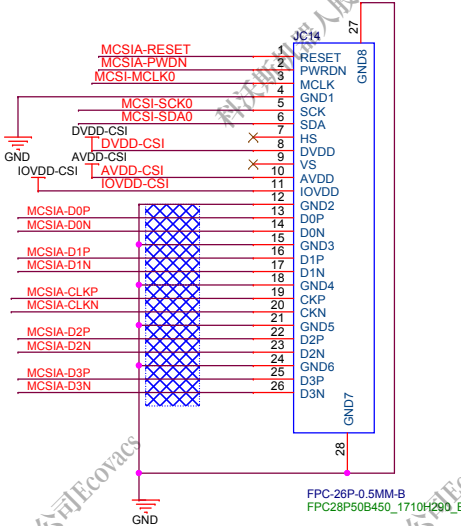


预置100mA 防止大硬盘插入 瞬间 系统 冲电 问题!

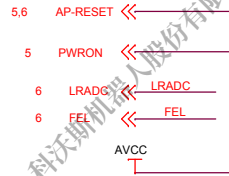
MIPI-CSI



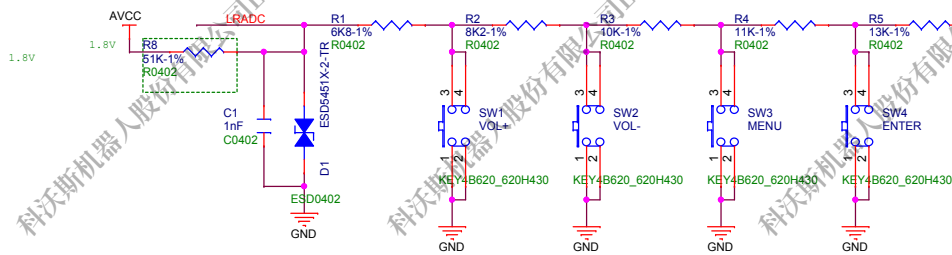
CAMERA A



KEY



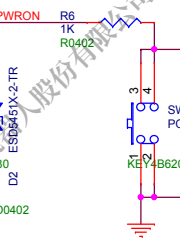
LRADCKEY



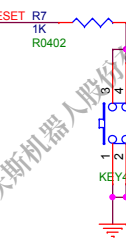
FEL



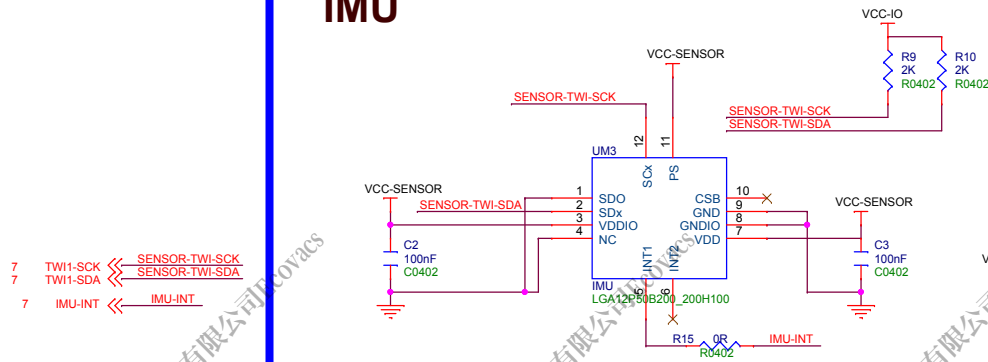
POWERON



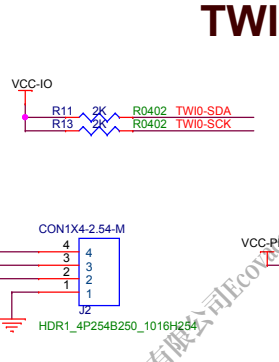
RESET



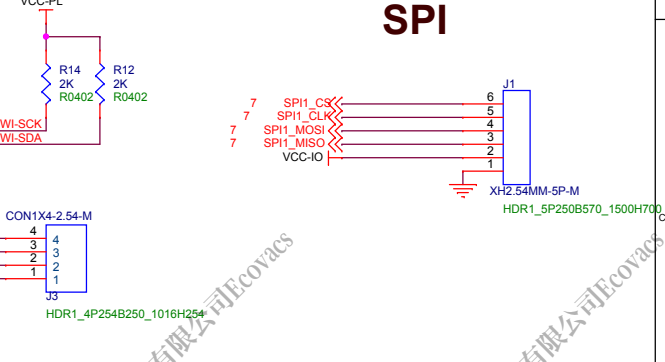
IMU



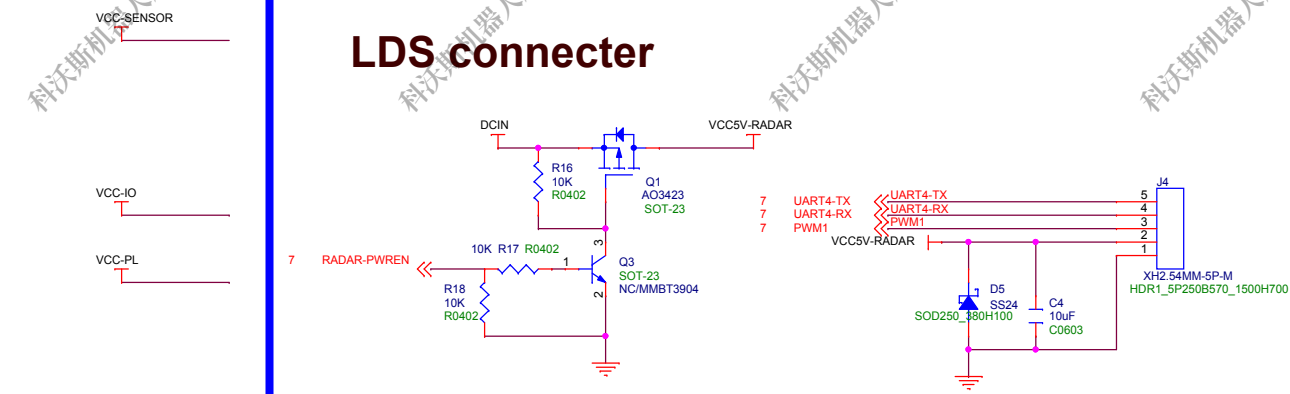
TWI



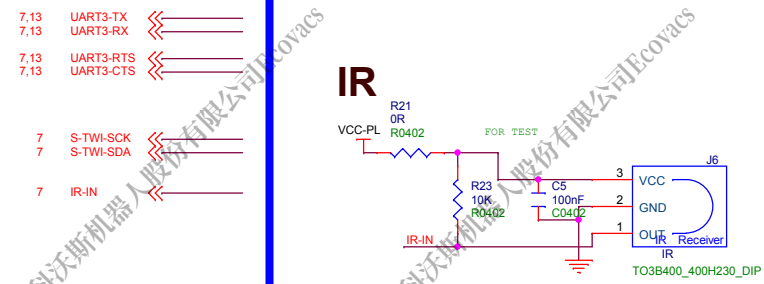
SPI



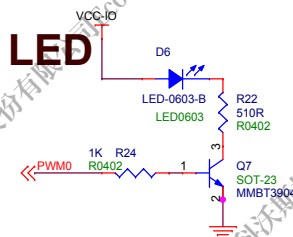
LDS connector



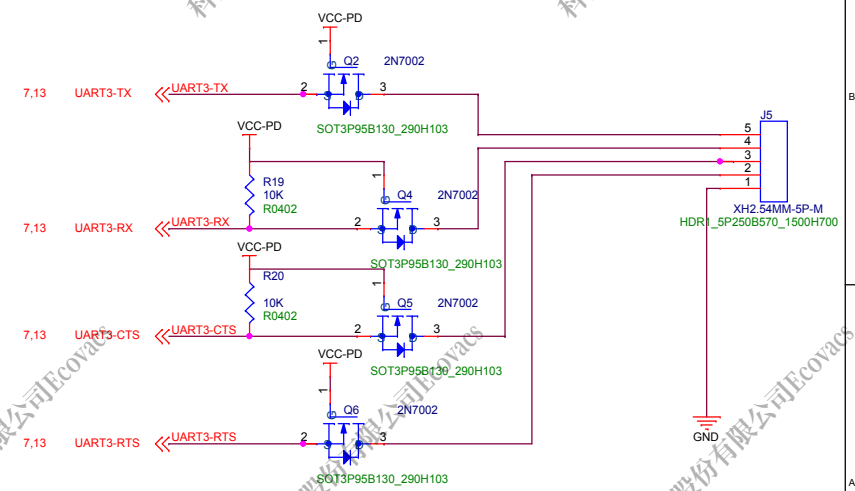
IR



LED



MCU connector



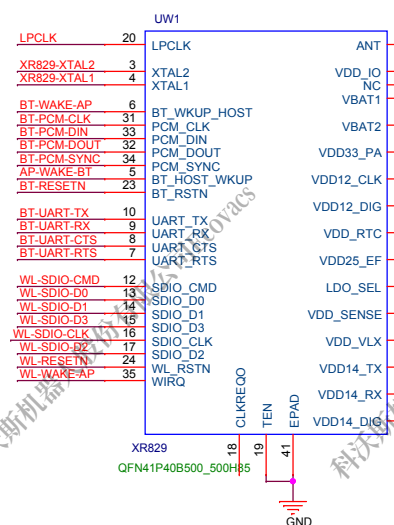
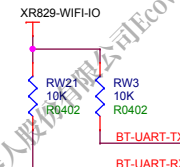
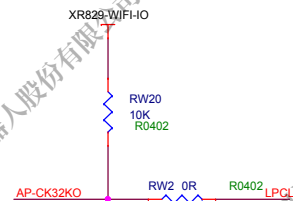
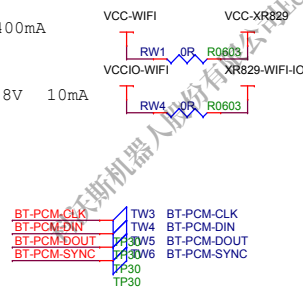
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WIFI XR829

7	WL-SDIO-CMD	WL-SDIO-CMD
7	WL-SDIO-D0	WL-SDIO-D0
7	WL-SDIO-D1	WL-SDIO-D1
7	WL-SDIO-D2	WL-SDIO-D2
7	WL-SDIO-D3	WL-SDIO-D3
7	WL-SDIO-CLK	WL-SDIO-CLK
7,11	WL-PMU-EN	WL-RESETN
7,11	WL-WAKE-AP	WL-WAKE-AP
7,11	BT-WAKE-AP	BT-WAKE-AP
7	BT-PCM-CLK	BT-PCM-CLK
7	BT-PCM-DIN	BT-PCM-DIN
7	BT-PCM-DOUT	BT-PCM-DOUT
7	BT-PCM-SYNC	BT-PCM-SYNC
7,11	AP-WAKE-BT	AP-WAKE-BT
7,11	BT-RST-N	BT-RESETN
7	BT-UART-TX	BT-UART-TX
7	BT-UART-RX	BT-UART-RX
7	BT-UART-CTS	BT-UART-CTS
6	AP-CK32K-OUT	AP-CK32K-OUT

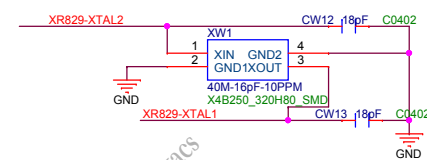
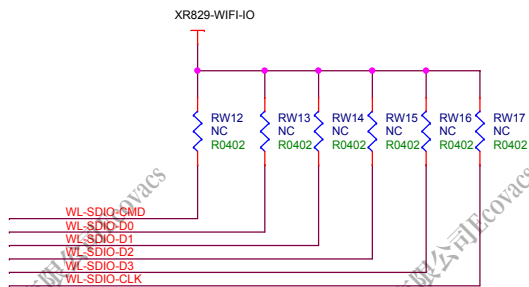
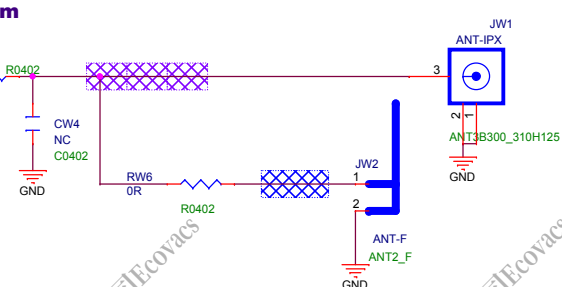
3.3V 400mA

3.3/1.8V 10mA



Place capacitors near the IC

50 ohm



DDR4 16X2

